CMP-4008Y Coursework 2 - Arcade System and Simulation

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Contents

Simulation.java

```
import java.io.*;
import java.util.*;

public class Simulation {

    public static Arcade initialiseArcade(String arcadeName,File gamesFile,File
        customerFile) throws FileNotFoundException
    {
        Arcade arcade = new Arcade(arcadeName);
        BufferedReader cReader = new BufferedReader(new FileReader(customerFile));
        BufferedReader gReader = new BufferedReader(new FileReader(gamesFile));
        String str;
        int line = 1;
        try
        {
            while((str = cReader.readLine()) != null)// while loop used to go throuch
                 each row and read them
            {
                String[] customerInfo = str.split("#");//splitting row by using # as
                    regex

                String id = customerInfo[0].trim();
                String name = customerInfo[1].trim();
                int balance = Integer.parseInt(customerInfo[2].trim());
                int age = Integer.parseInt(customerInfo[3].trim());
                int numbOfElements = 4;//this variable is needed to check how many
                    elements does array contain if it is equal to this variable
                // then it means there is level of discount missing which means it is
                     none
                if(numbOfElements == customerInfo.length)
                {
                    LevelOfDiscount discountLVL = LevelOfDiscount.NONE;
                    Customer customer = new Customer(id,name,balance,age,discountLVL)
                        ;
                    arcade.addCustomer(customer);
                }
                else
                {
                    String discountString = customerInfo[4].trim();
                    LevelOfDiscount discountLVL = LevelOfDiscount.valueOf(
                        discountString);
                    Customer customer = new Customer(id,name,balance,age,discountLVL)
                        ;
                    arcade.addCustomer(customer);
                }

                line++;
            }

            while ((str = gReader.readLine()) != null)
            {
                String[] gamesInfo = str.split("@");

                String id = gamesInfo[0].trim();
                String name = gamesInfo[1].trim();
                String type = gamesInfo[2].trim();
                int price = Integer.parseInt(gamesInfo[3].trim());
                if(type.equals("cabinet"))
                {
                    String reward = gamesInfo[4].trim();
                    if(reward.equals("yes"))
```

```
                    {
                        boolean hasReward = true;
                        CabinetGame game = new CabinetGame(name,id,price,hasReward);
                        arcade.addArcadeGame(game);
                    }
                    else
                    {
                        boolean hasReward = false;
                        CabinetGame game = new CabinetGame(name,id,price,hasReward);
                        arcade.addArcadeGame(game);
                    }

                }
                else if(type.equals("active"))
                {
                    int ageLimit = Integer.parseInt(gamesInfo[4].trim());
                    ActiveGame game = new ActiveGame(name,id,price,ageLimit);
                    arcade.addArcadeGame(game);
                }
                else
                {
                    int ageLimit = Integer.parseInt(gamesInfo[4].trim());
                    String equipment = gamesInfo[5];
                    if(equipment.equals("headsetAndController"))
                    {
                        Equipment equipmentType = Equipment.HEADSET_AND_CONTROLLER;
                        VirtualRealityGame game = new VirtualRealityGame(name,id,
                            price,ageLimit,equipmentType);
                        arcade.addArcadeGame(game);
                    }
                    else if(equipment.equals("headsetOnly"))
                    {
                        Equipment equipmentType = Equipment.HEADSET_ONLY;
                        VirtualRealityGame game =  new VirtualRealityGame(name,id,
                            price,ageLimit,equipmentType);
                        arcade.addArcadeGame(game);
                    }
                    else
                    {
                        Equipment equipmentType = Equipment.BODY_TRACKING_SUIT;
                        VirtualRealityGame game = new VirtualRealityGame(name,id,
                            price,ageLimit,equipmentType);
                        arcade.addArcadeGame(game);
                    }
                }

            }
        }
        catch (IOException e)
        {
            e.printStackTrace();
        }
        return arcade;
    }

    public static void simulateFun(Arcade arcade ,File tranFile) throws
        FileNotFoundException
    {

        BufferedReader tReader = new BufferedReader(new FileReader(tranFile));
        String str;
        int line = 1;
```

```
113         try
            {
115             while((str = tReader.readLine()) != null)//while loop is used to check
                    every single row and it has been divided on three branching parts
                    where first one checks and performs transaction to allow customer to
                    play
                {
117

119                 String[] transactions = str.split(",");//splitting each row on
                        certain amount of elements based on row itself
                    String action = transactions[0].trim();
121                 String custID = transactions[1].trim();
                    if(action.equals("PLAY"))
123                 {
                        try
125                     {
                            String gameID = transactions[2].trim();
127                         String peakStatus = transactions[3].trim();
                            boolean peak;
129                         if(peakStatus.equals("PEAK"))
                            {
131                             peak = true;
                            }
133                         else
                            {
135                             peak = false;
                            }
137
                            boolean result = arcade.processTransaction(custID,gameID,peak
                                );
139                         if(result)
                            {
141                             String getName = arcade.getCustomer(custID).
                                    getCustomerName();
                                System.out.println(getName + ", you have successfully
                                    paid for the game. Have fun!");
143                         }

145
                        }
147                     catch (IncorrectIdException e)
                        {
149                         System.out.println("Wrong format of id: " + e.getMessage());
                        }
151

153                 }
                    else if (action.equals("NEW_CUSTOMER"))//"The second part is about
                        identifying the type of row to decide where a new customer should
                         be added to the system."
155                 {
                        String name = transactions[2].trim();
157                     int numbOfElements = 6;//same principle used as in
                            initialiseArcade method
                        if(numbOfElements == transactions.length)
159                     {
                            String discountLVL = transactions[3].trim();
161
                            LevelOfDiscount level = LevelOfDiscount.valueOf(discountLVL);
163                         int balance = Integer.parseInt(transactions[4].trim());
                            int age = Integer.parseInt(transactions[5].trim());
165                         Customer customer = new Customer(custID,name,balance,age,
```

```
                            level);
                        arcade.addCustomer(customer);
                        checkForSuccess(arcade,custID);
                    }
                    else
                    {
                        int balance = Integer.parseInt(transactions[3].trim());
                        int age = Integer.parseInt(transactions[4].trim());
                        LevelOfDiscount discountLVL = LevelOfDiscount.NONE;
                        Customer customer = new Customer(custID,name,balance,age,
                            discountLVL);
                        arcade.addCustomer(customer);
                        checkForSuccess(arcade,custID);
                    }
                }
                else //And if first element of the array starts with ADD_FUNDs this
                    parts performing adding money to the customer balance
                {
                    int fee = Integer.parseInt(transactions[2].trim());
                    Customer customer = arcade.getCustomer(custID);
                    try
                    {
                        int oldBalance = customer.getAccBalance();
                        customer.addFunds(fee);
                        if(oldBalance < customer.getAccBalance() )
                        {
                            System.out.println("You have successfully added money to
                                your balance");
                        }
                        else
                        {
                            System.out.println("Money haven't been added,ask for
                                assistance");
                        }
                    }
                    catch (NullPointerException e)
                    {
                        System.out.println("Failed to add funds");
                    }


                }
                line++;
            }
            System.out.println( "\n" + "The richest customer is " +  arcade.
                findRichestCustomer());
            System.out.println("\n"+  "The median game price is: " + arcade.
                getMedianGamePrice());
            System.out.println("\n"+  "Cabinet/Active/Virtual games: " + Arrays.
                toString(arcade.countArcadeGames()));
            Arcade.printCorporateJargon();
            System.out.println();


        }
        catch (IOException e)
        {
            e.printStackTrace();
        }


    }
```

```
221     public static void checkForSuccess(Arcade arcade, String custID) // I have
            created this method just to avoid repetition of the code.
        // This method checks whether adding customer to the customers ArrayList is
            successful or not
223     {
            boolean existInsystem = false;
225         for (int i = 0; i < arcade.customers.size();i++)
            {
227             if(arcade.customers.get(i).getCustomerID().equals(custID))//Loop through
                    the array of customers until custID is equal to the id of one of the
                    customers
                {
229                 existInsystem = true;
                }
231         }
            if(existInsystem)
233         {
                System.out.println("Customer has successfully been added to the system");
235         }
            else
237         {
                System.out.println("Customer haven't been added,ask for assistance");
239         }
        }
241
        public static void main(String[] args)
243     {
            File custFile = new File("customers.txt");
245         File gameFile =new File("games.txt");
            File trFile = new File("transactions.txt");
247         try
            {
249             Arcade passedArcade = initialiseArcade("Arcade of fun ",gameFile,custFile
                    );
                simulateFun(passedArcade,trFile);
251             System.out.println(passedArcade);


253
            }
255         catch (IOException e)
            {
257             e.printStackTrace();
            }
259     }
    }
```

ArcadeGame.java

```
 abstract public class ArcadeGame implements Comparable<ArcadeGame>
{
    protected String name;
    protected String id;
    protected int price;
    protected double newPrice;

    public ArcadeGame(String name,String id, int price)
    {
        this.name = name;
        this.id = id;
        this.price = price;
        if(id.length() != 10) //Checking whether id is 10 characters long
        {
            throw new IncorrectIdException("Id should be 10 characters long!");
        }
    }

    public String getName()
    {
        return name;
    }
    public String getID()
    {
        return id;
    }
    public int getPrice()
    {
        return price;
    }
    public int compareTo(ArcadeGame other)
    {
        return Integer.compare(this.price,other.price);//The compareTo method is used
            to perform sorting on an ArrayList of games
    }


    public abstract int calculatePrice(boolean peak);

//    public String toString()
//    {
//        return "The " + name + " has id: " + id + " and its price is " + price;
//    }

//    public static void main(String[] args)
//    {
//        ArcadeGame obj = new ArcadeGame("race","1003238488",1);
//        System.out.println(obj);
//        ArcadeGame obj1 = new ArcadeGame("race","100323848",1);
//
//    }
    //The main method is commented out because i used it for testing to check whether
        the exception is thrown
    // and whether the price is shown in pence
}
```

CabinetGame.java

```
public class CabinetGame extends ArcadeGame {
    private boolean hasReward;

    public CabinetGame(String name,String id,int price,boolean hasReward)
    {
        super(name,id,price);
        this.hasReward = hasReward;
        if(!id.startsWith("C"))
        {
            throw new IncorrectIdException("The id should start from the letter C");
        }
    }
    public boolean getHasReward()
    {
        return hasReward;
    }
    public int calculatePrice(boolean peak)//overridden calculatePrice method
    {
        if(peak == true)// if it is peak time then price stays the same
        {
            newPrice = price;
            return (int) newPrice;
        }
        else if (peak == false && hasReward == false)//if it is not peak time and
            there is no reward price gets 50% discount
        {
            newPrice = price * 0.5;
            return (int) newPrice;
        }
        else if (peak == false && hasReward == true)//This time if peak time and
            there is reward price gets 20% discount
        {
            newPrice = price * 0.8;
            return (int) newPrice;
        }
        else//in other cases price stays the same
        {
            newPrice = price;
            return (int) newPrice;
        }

    }
    public String toString()
    {
        return "\n"+ "\n" + "Name: " + name + "\n" + "Id: " + id + "\n" + "Price: " +
             price + "p \n" + "Status of the reward: : " + hasReward;
    }

//    public static void main(String[] args)
//    {
////        CabinetGame cabGame = new CabinetGame("Pac-Man","C003322111",6,true);
////        System.out.println(cabGame.calculatePrice(false));
////        System.out.println(cabGame);
//        CabinetGame cabinetGame = new CabinetGame("aafaf","C000111222",500,true);
//        System.out.println(cabinetGame.toString());
//    }

    //Exception works well,if first character of id is not a letter C then exception
        appears.The same with number of characters.
    // Also,price is returned properly according to the discounts
```

```
58 }
```

ActiveGame.java

```
public class ActiveGame extends ArcadeGame {
    protected int minAge;

    public ActiveGame(String name,String id,int price,int minAge)
    {
        super(name,id,price);
        this.minAge = minAge;
        if(id.length() != 10 || !id.startsWith("A"))
        {
            throw new IncorrectIdException("The id should start from the letter A
               because it is Active Game");
        }
    }
    public int getMinAge()
    {
        return minAge;
    }
    public int calculatePrice(boolean peak) //same principle used as in Cabin game
       but without 50% discount and without rewards regardless age limit
    {
        if(peak == true)
        {
            newPrice = price;
            return (int) newPrice;
        }
        else if (peak == false)
        {
            newPrice = price * 0.8;
            return (int) newPrice;
        }
        else
        {
            newPrice = price;
            return (int) newPrice;
        }

    }
    public String toString()
    {
//        return "Price of the game " + name + " with id " + id + " is: " + (int)
   newPrice + "p and minimum age to play the game is: " + minAge;
        return  "\n"+ "\n" + "Name: " + name + "\n" + "Id: " + id + "\n" + "Price: "
           + price + "p \n" + "Minimum age:" + minAge;
    }

//    public static void main(String[] args)
//    {
//        ActiveGame actGame = new ActiveGame("Air Hockey","A003322111",5,12);
//        System.out.println(actGame.calculatePrice(true));
//        System.out.println(actGame);
//    }

    //I have tested the ActiveGame class the same way as cabinetGame class
}
```

VirtualRealityGame.java

```
public class VirtualRealityGame extends ActiveGame{
//    public enum Equipment {HEADSET_ONLY,HEADSET_AND_CONTROLLER,BODY_TRACKING_SUIT};
    private Equipment vrEquipment;

    public VirtualRealityGame (String name,String id,int price,int minAge,Equipment
        vrEquipment)
    {
        super(name,id,price,minAge);
        this.vrEquipment = vrEquipment;
        if(id.length() != 10 || !id.startsWith("AV"))
        {
            throw new IncorrectIdException("The id should start from the letter AV
                because it is Active Virtual Game");
        }

    }
    public Equipment getVrEquipment()
    {
        return vrEquipment;
    }

    public int calculatePrice(boolean peak)//checks peak time and type of Equipment
    {
        if(peak == true)
        {
            newPrice = price;
            return (int) newPrice;
        }
        else if (peak == false && vrEquipment == Equipment.HEADSET_ONLY)
        {
            newPrice = price * 0.9;
            return (int) newPrice;
        }
        else if (peak == false && vrEquipment == Equipment.HEADSET_AND_CONTROLLER)
        {
            newPrice = price * 0.95;
            return (int) newPrice;
        }
        else
        {
            newPrice = price;
            return (int) newPrice;
        }
    }
    public String toString()
    {
//        return  "Price of the game " + name + " with id " + id + " is: " + (int)
    newPrice + "p and equipment needed for this game: " + vrEquipment;
        return  "\n"+ "\n" + "Name: "+ name + "\n" + "Id: " + id + "\n" + "Price: " +
            price + "p \n" + "Equipment type: " + vrEquipment;
    }

//    public static void main(String[] args)
//    {
//        //Testing whether everything works fine
//        VirtualRealityGame vrgame = new VirtualRealityGame("VR boxing","AV00000000
    ",4,10,Equipment.BODY_TRACKING_SUIT);
//        VirtualRealityGame vrgame = new VirtualRealityGame("VR boxing","AV00000000
    ",4,10,Equipment.HEADSET_ONLY);
//        //Testing whether discounts work fine
//        System.out.println(vrgame.calculatePrice(true));
```

```
56 //         System.out.println(vrgame.calculatePrice(false));
   //         System.out.println(vrgame);
58 //
   //       //Testing whether there is sn exception related to the first letter
60 //       VirtualRealityGame vrgame1 = new VirtualRealityGame("VR boxing
       ","0000000000",4,10,Equipment.BODY_TRACKING_SUIT);
   //       //Testing related to the first two letters of the id because of VRGame class
62 //       VirtualRealityGame vrgame2 = new VirtualRealityGame("VR boxing","A000000000
       ",4,10,Equipment.BODY_TRACKING_SUIT);
   //      //Testing related to the number of characters
64 //       VirtualRealityGame vrgame3 = new VirtualRealityGame("VR boxing","AV000000000
       ",4,10,Equipment.BODY_TRACKING_SUIT);
   //    }
66

68 }
```

Customer.java

```
public class Customer implements Comparable<Customer>{
    private String customerID;
    private String customerName;
    private int age;
    private LevelOfDiscount levelOfDiscount;
    private int accBalance;

    public Customer(String customerID,String customerName,int age,LevelOfDiscount
        levelOfDiscount)
    {
        this.customerID = customerID;
        if(customerID.length() != 6)
        {
            throw  new InvalidCustomerException("Id should contain 6 characters!");
        }
        this.customerName = customerName;
        this.age = age;
        this.levelOfDiscount = levelOfDiscount;
        accBalance = 0;
    }

    public Customer (String customerID,String customerName,int accBalance,int age,
        LevelOfDiscount levelOfDiscount)
    {
        this(customerID,customerName,age,levelOfDiscount);
        if(levelOfDiscount != LevelOfDiscount.STUDENT && accBalance < 0)//throws
            exception if balance goes less than zero if it is not student
        {
            throw new InvalidCustomerException("Account balance can't be zero!");
        }
        else if(customerID.length() != 6)//checks whether id has 6 characters
        {
            throw  new InvalidCustomerException("Id should contain 6 characters!");
        }
        else {
            this.accBalance = accBalance;
        }
        if(levelOfDiscount == LevelOfDiscount.STUDENT)//allows balance to go lower
            than zero if it is a student
        {
            if(accBalance <-500)
            {
                throw new InvalidCustomerException("Student's balance cannot be lower
                    than -500!");
            }
            else
            {
                this.accBalance = accBalance;
            }
            this.accBalance = accBalance;
        }

    }

    public String getCustomerID()
    {
        return customerID;
    }
    public String getCustomerName()
    {
        return customerName;
```

```
    }
58  public int getAge()
    {
60      return age;
    }
62  public LevelOfDiscount getLevelOfDiscount()
    {
64      return levelOfDiscount;
    }
66  public int getAccBalance()
    {
68      return accBalance;
    }
70

72  public int addFunds(int amount)//adds money only if positive amount is entered
    {
74      if(amount<=0)
        {
76          System.out.println("Please enter the positive amount!");

78      }
        else
80      {
            return accBalance = accBalance + (amount);
82      }
        return accBalance;
84  }

86  public int chargeAccount(ArcadeGame game,boolean isPeak)//checked exception is
        thrown if balance doesn't have enough credits
    {
88      double totalCharge;

90     if(accBalance < game.getPrice())//
       {
92          try
            {
94              throw new InsufficientBalanceException("Your balance does not have
                    enough credits");
            }
96          catch (InsufficientBalanceException e)
            {
98              System.out.println( e.getMessage());
                return 0;
100         }

102    }
       if(game instanceof ActiveGame)//Also checks and throws checked exception if
           customer tries to play ActiveGame but they are too young
104    {
           if(age < ((ActiveGame) game).minAge)
106        {
               try
108            {
                   throw new AgeLimitException("You are too young!");
110            }
               catch (AgeLimitException e)
112            {
                   System.out.println( e.getMessage());
114                return 0;
               }
116
```

```
            }
118
        }
120     if(levelOfDiscount == LevelOfDiscount.STUDENT)//Gives discount based ong the
            level of discount
        {
122         totalCharge = game.calculatePrice(isPeak) * 0.95;
            accBalance = accBalance - (int)totalCharge;
124         return (int)totalCharge;
        }
126     else if(levelOfDiscount == LevelOfDiscount.STAFF)
        {
128         totalCharge = game.calculatePrice(isPeak) * 0.90;
            accBalance = accBalance - (int)totalCharge;
130         return (int)totalCharge;
        }
132     else
        {
134         totalCharge = game.calculatePrice(isPeak);
            accBalance = accBalance - (int)totalCharge;
136         return (int)totalCharge;
        }
138

140 }

142 public int compareTo(Customer other)
    {
144     return Integer.compare(this.accBalance,other.accBalance);//Used to sort
            ArrayList of customers to determoine the richest
    }
146

148 public String toString()
    {
150
        return  "\n"+ "\n" + "Name: " + customerName + "\n" + "Id: " + customerID +
            "\n" + "Age: " + age + "\n" + "Level of discount: " + levelOfDiscount
          + "\n" + "Current balance: " + accBalance;
152 }
//    public static void main(String[] args)
154 //    {
////        Customer customer = new Customer("123456","Alex",14,LevelOfDiscount.NONE
    ,5);
156 //        //chargeAccount methd works well when the levevel of the discount is none
////        Customer customer = new Customer("123456","Alex",14,LevelOfDiscount.
    STUDENT,5);
158 //        //chargeAccount works well when level of the discount is STUDENT
////        Customer customer = new Customer("123456","Alex",14,LevelOfDiscount.
    CMP_STAFF,5);
160 //        //The same with CMP_STAFF
////        Customer customer = new Customer("123456","Alex",8,LevelOfDiscount.
    CMP_STAFF,5);
162 //        //Method has successfully given AgeLimitException
////        Customer customer = new Customer("123456","Alex",14,LevelOfDiscount.
    CMP_STAFF,1);
164 //        //Method has successfully  given InsuficientBalanceException
//        ActiveGame actGame = new ActiveGame("Game","A002211345",2,10);
166 //        System.out.println(customer.chargeAccount(actGame,false));
//
168 // //        addFunds method succesfully adds funds
////        System.out.println(customer);
170 ////        int newBalance = customer.addFunds(1);
```

```
////        System.out.println(newBalance);

//    Customer cusstomer = new Customer("C45678","Adam",12,LevelOfDiscount.STUDENT);
    toString testing
//    System.out.println(cusstomer.toString());
//
//    }
//
}
```

Arcade.java

```
2  import java.util.ArrayList;
   import java.util.Collections;
4
   public class Arcade {
6      private String arcadeName;
       private int revenue;
8      ArrayList<ArcadeGame> arcadeGamesCollection = new ArrayList<>();
       ArrayList<Customer> customers = new ArrayList<>();
10
       public Arcade (String arcadeName)
12     {
           this.arcadeName = arcadeName;
14     }

16     public void addCustomer(Customer customer)
       {
18         customers.add(customer);
       }
20
       public void addArcadeGame(ArcadeGame game)
22     {
           arcadeGamesCollection.add(game);
24     }

26     public String getArcadeName()
       {
28         return arcadeName;
       }
30
       public int getRevenue()
32     {
           return revenue;
34     }


36
       public Customer getCustomer(String customerID) throws InvalidCustomerException
38     //This method iterates through all clients and, if the provided id matches,
           returns the correct customer
       {
40         if(customerID.length() !=6 )
           {
42             throw new InvalidCustomerException("Id should have 6 characters");
           }
44         for(int i = 0; i<customers.size();i++)
           {
46             if(customers.get(i).getCustomerID().equals(customerID))
               {
48                     return customers.get(i);
               }
50         }
           System.out.println("There is no customer with provided id");
52        return null;

54     }

56     public ArcadeGame getArcadeGame(String gameID) throws IncorrectIdException
       //This method iterates through all games and, if the provided id matches, returns
            the correct game
58     {
           boolean exist = false;
```

```
60          for(int i=0; i<arcadeGamesCollection.size(); i++)
            {
62              if(arcadeGamesCollection.get(i).getID().equals(gameID))
                {
64                  exist = true;
                    return arcadeGamesCollection.get(i);
66
                }
68              if(gameID.length() !=10 && (!gameID.startsWith("C") || !gameID.startsWith
                    ("A")))
                {
70
                    throw new IncorrectIdException("id should contain 10 characters and
                        start from letter A or C depends on the game!");
72              }
            }
74          if(!exist)
            {
76              System.out.println("There is no game with that id");
            }
78
            return null;
80      }

82      public boolean processTransaction(String customerID,String gameID,boolean peak)
        //this method gets game andd customer by their id and checks what type of the
            game is that
84      {
            boolean wasUpdated = false;
86          Customer customer = getCustomer(customerID);
            ArcadeGame game = getArcadeGame(gameID);
88          int balance = customer.getAccBalance();
            int price = game.getPrice();
90          if(game instanceof ActiveGame)//if it is ActiveGame then it checks minimum
                age and balance
            {
92              if(balance >= price && customer.getAge() >= ((ActiveGame) game).minAge)
                {
94
                    int charge = customer.chargeAccount(game,peak);
96                  revenue = revenue + charge;
                    wasUpdated = true;
98
                }
100             else
                {
102                 if(customer.getAge() < ((ActiveGame) game).minAge)
                    {
104                     try
                        {
106                         throw new AgeLimitException("You are too young");
                        }
108                     catch (AgeLimitException e)
                        {
110                         System.out.println(e.getMessage());

112                     }
                    }
114                 if(balance < price)
                    {
116                     try
                        {
118                         String getName = getCustomer(customerID).getCustomerName();
```

```
                        throw new InsufficientBalanceException(getName + ", you don't
                            have enough credits");
                    }
                    catch (InsufficientBalanceException e)
                    {
                        System.out.println(e.getMessage());
                    }
                }


            }
        }
        else//If it is Cabinet game then it checks just a balance
        {
            if(balance >= price)
            {
                int charge = customer.chargeAccount(game,peak);
                revenue = revenue + charge;
                wasUpdated = true;
            }
            else
            {
                try
                {
                    String getName = getCustomer(customerID).getCustomerName();
                    throw new InsufficientBalanceException(getName + ", you don't
                        have enough credits");
                }
                catch (InsufficientBalanceException e)
                {
                    System.out.println(e.getMessage());
                }
            }
        }

        return wasUpdated;
    }

    public Customer findRichestCustomer()
    {
        Collections.sort(customers);
//        for(int i =0;i<customers.size();i++)
//        {
//            System.out.println(customers.get(i)); Testing whether ArrayList is
    sorted
//        }

        Customer richest = customers.get(0);//checks and compares each customer with
            the first one and assigns to the variable richest until the end of the
            ArrayList
        for(int i = 1;i<customers.size() ;i++)
        {
            if(customers.get(i).getAccBalance() > richest.getAccBalance())
            {
                richest = customers.get(i);

            }

        }
        return richest;

    }

```

```
178     public int getMedianGamePrice()
        {
180         Collections.sort(arcadeGamesCollection);

182         if(arcadeGamesCollection.size() % 2 == 0)//checks whether ArrayList has even
                number of games if it is even then it finds two median ones and divides
                them by 2
            {
184             int numberOfElemnts = arcadeGamesCollection.size();
                ArcadeGame median1 = arcadeGamesCollection.get(numberOfElemnts/2);
186             ArcadeGame median2 = arcadeGamesCollection.get(numberOfElemnts/2 -1);
                int meddian = (median1.getPrice() + median2.getPrice()) / 2;
188             return meddian;
            }
190         else//if it odd number of games then just find median
            {
192             ArcadeGame medianElement = arcadeGamesCollection.get(
                    arcadeGamesCollection.size() / 2);
                int median = medianElement.getPrice();
194             return median;
            }
196     }


198
        public  int[] countArcadeGames()
200     {//This method simply checks the current type of each game and increments the
            count for that type until the end of the ArrayList is reached.
            int[] arcadeGames = new int[3];
202         int cabinetG = 0;
            int activeG = 0;
204         int vrG = 0;
            for(int i=0;i<arcadeGamesCollection.size(); i++)
206         {
                if(arcadeGamesCollection.get(i) instanceof CabinetGame)
208             {
                    cabinetG++;
210                 arcadeGames[0] = cabinetG;
                }
212             else if(arcadeGamesCollection.get(i) instanceof VirtualRealityGame)
                {
214                 vrG++;
                    arcadeGames[2] = vrG;
216             }
                else
218             {
                    activeG++;
220                 arcadeGames[1] = activeG;
                }
222             if(i == arcadeGamesCollection.size() -1)
                {
224 //                  System.out.println(arcadeGames[2]); used for testing to check how
        many games are there
    //                  System.out.println(arcadeGames[0]);
226
                    return arcadeGames;
228             }

230         }

232         return arcadeGames;
        }
234
```

```
    public static void printCorporateJargon()
    {
        System.out.println("GamesCo does not take responsibility for any accidents or
            fits of rage that occur on the premises");
    }

    public String toString()
    {
        return "Arcade name: " + arcadeName + "\n" + "Revenue:  "+ revenue +"\n"+ "\n
            " + "Games list: " + arcadeGamesCollection + "\n" + "\n" + "List of
            customers: " + customers;
    }


//    public static void main(String[] args)
//    {
//////
//            Customer customer = new Customer("100000","Bob",600,10, LevelOfDiscount
    .STAFF);
//            ActiveGame game = new ActiveGame("great","A100222333",100,10);
//            CabinetGame game1 = new CabinetGame("sss","C100200300",500,true);
//            VirtualRealityGame game2 = new VirtualRealityGame("aaa","AV10020033
    ",400,10, Equipment.BODY_TRACKING_SUIT);
//////        VirtualRealityGame game3 = new VirtualRealityGame("aava","AV10020033
    ",6,12, VirtualRealityGame.Equipment.BODY_TRACKING_SUIT);
//////        VirtualRealityGame game4 = new VirtualRealityGame("aaab","AV10020033
    ",9,12, VirtualRealityGame.Equipment.BODY_TRACKING_SUIT);
//////        VirtualRealityGame game5 = new VirtualRealityGame("aadab","AV10020033
    ",10,12, VirtualRealityGame.Equipment.BODY_TRACKING_SUIT);
//            Arcade arcade = new Arcade("Tennis");
//            arcade.addArcadeGame(game);
//            arcade.addArcadeGame(game1);
//            arcade.addArcadeGame(game2);
//////        arcade.addArcadeGame(game3);
//////        arcade.addArcadeGame(game4);
//////        arcade.addArcadeGame(game5);
//////        arcade.getMedianGamePrice(); // Testing of the getMedianGamePrice
    succesfully sorts array base omn the price and returns median
////
//////        arcade.countArcadeGames();//Testing of the countArcadeGames ,it counts
    precise number of games and separate ActiveGames and VirtualRealityGames
////
////
//            arcade.addCustomer(customer);
//////        arcade.addArcadeGame(game); Testing of the addGame
////
//////        Testing of the proccessTransaction.succesfully returns true if
    tracnsaction completed,false if not and stops transaction based on the exceptions
//            Boolean transactionStatus = arcade.processTransaction(customer.
    getCustomerID(),game1.getID(),true);
//            System.out.println("Transaction status is : " + transactionStatus +"
    and current balance of the customer is " + customer.getAccBalance());
////////        arcade.customers.add(customer);
////////        Customer str = arcade.getCustomer("100001");
////////        System.out.println(str);
////
////
//////       Testing whether findRichestCustomer displays actually the richest and it
     does
//////        Arcade arcade = new Arcade("Tennis");
//////        Customer c1 = new Customer("123456","Alex",1,20, Customer.
    LevelOfDiscount.CMP_STAFF);
```

```
//////        Customer c2 = new Customer("123458","John",2,21, Customer.
    LevelOfDiscount.CMP_STAFF);
//////        Customer c3 = new Customer("123457","Bob",3,22, Customer.
    LevelOfDiscount.CMP_STAFF);
//////        Customer c4= new Customer("123457","Sara",3,22, Customer.
    LevelOfDiscount.CMP_STAFF);
//////        Customer c5 = new Customer("123457","Jess",10,22, Customer.
    LevelOfDiscount.CMP_STAFF);
//////        arcade.addCustomer(c1);
//////        arcade.addCustomer(c2);
//////        arcade.addCustomer(c3);
//////        arcade.addCustomer(c4);
//////        arcade.addCustomer(c5);
//////        arcade.findRichestCustomer();
//
//
//    }
}
```

AgeLimitException.java

```
public class AgeLimitException extends Exception {
    public AgeLimitException(String message) {
        super(message);
    }
}
```

Equipment.java

```
1 public enum Equipment {
      HEADSET_ONLY,HEADSET_AND_CONTROLLER,BODY_TRACKING_SUIT
3 }
```

IncorrectIdException.java

```
public class IncorrectIdException extends RuntimeException {
    public IncorrectIdException(String message)
    {
        super(message);
    }
}

// If I need to throw exception inside the constructor then it means
// I need to extend RuntimeException instead of Exception class because it should be
    unchecked exception
```

InsufficientBalanceException.java

```
1 public class InsufficientBalanceException extends Exception {
      public InsufficientBalanceException(String message) {
3         super(message);
      }
5 }
```

InvalidCustomerException.java

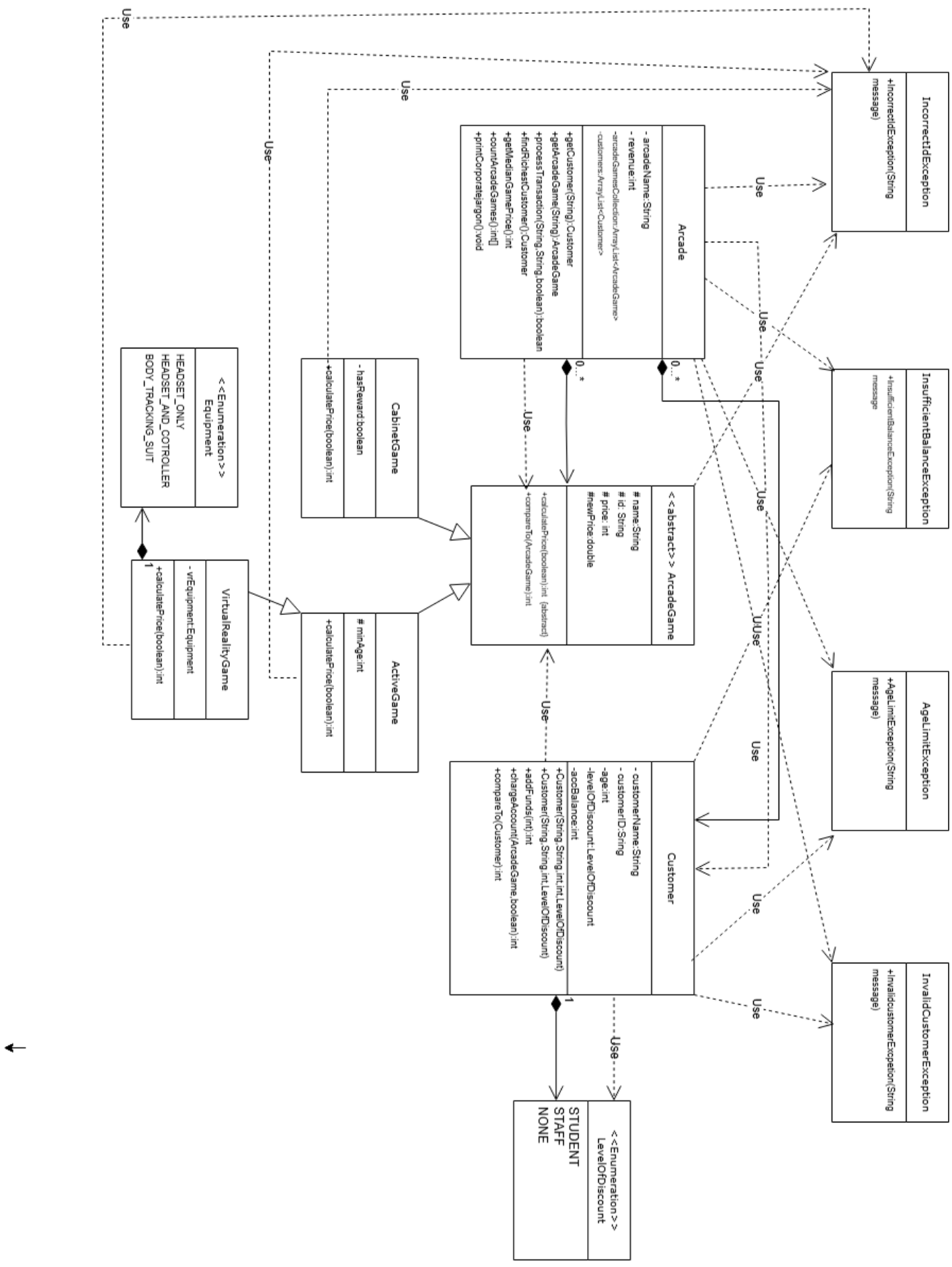
```
1 public class InvalidCustomerException extends RuntimeException {
      public InvalidCustomerException(String message)
3     {
          super(message);
5     }
  }
```

LevelOfDiscount.java

```
public enum LevelOfDiscount
{
    NONE,STAFF,STUDENT
}
```

diagram.pdf

Binary file included. Size: 83,565 bytes.
(Included PDF starts on next page.)


IncorrectIdException
+IncorrectIdException(String message)
InsufficientBalanceException
+InsufficientBalanceException(String message)
AgeLimitException
+AgeLimitException(String message)
InvalidCustomerException
+InvalidcustomerException(String message)
Arcade
- arcadeName:String
- revenue:int
-arcadeGamesCollection:ArrayList<ArcadeGame>
-customers:ArrayList<Customer>
+getCustomer(String):Customer
+getArcadeGame(String):ArcadeGame
+processTransaction(String,String,boolean):boolean
+findRichestCustomer():Customer
+getMedianGamePrice():int
+countArcadeGames():int[]
+printCorporatejargon():void
<<abstract>> ArcadeGame
name:String
id: String
price: int
#newPrice:double
+calculatePrice(boolean):int {abstract}
+compareTo(ArcadeGame):int
CabinetGame
- hasReward:boolean
+calculatePrice(boolean):int
ActiveGame
minAge:int
+calculatePrice(boolean):int
VirtualRealityGame
- vrEquipment:Equipment
+calculatePrice(boolean):int
<<Enumeration>>
Equipment
HEADSET_ONLY
HEADSET_AND_COTROLLER
BODY_TRACKING_SUIT
Customer
- customerName:String
- customerID:String
-age:int
-levelOfDiscount:LevelOfDiscount
-accBalance:int
+Customer(String,String,int,int,LevelOfDiscount)
+Customer(String,String,int,LevelOfDiscount)
+addFunds(int):int
+chargeAccount(ArcadeGame,boolean):int
+compareTo(Customer):int
<<Enumeration>>
LevelOfDiscount
STUDENT
STAFF
NONE
Use
0...*
1

Application

Compiler Invocation

```
javac -Xlint:unchecked -Xlint:deprecation -encoding UTF-8 -d
    prepasg13494941047599502156classes Simulation.java ArcadeGame.java CabinetGame.
    java ActiveGame.java VirtualRealityGame.java Customer.java Arcade.java
    AgeLimitException.java Equipment.java IncorrectIdException.java
    InsufficientBalanceException.java InvalidCustomerException.java LevelOfDiscount.
    java
```

Compiler Messages

None.

Application Invocation

```
java Simulation
```

Messages to STDOUT

```
Amanda Pham, you have successfully paid for the game. Have fun!
Amber Y Basse, you have successfully paid for the game. Have fun!
Kiri Kramer, you have successfully paid for the game. Have fun!
Amanda Pham, you don't have enough credits
Charis Jaramillo, you have successfully paid for the game. Have fun!
Aman Crosby, you have successfully paid for the game. Have fun!
Aoife Crouch, you have successfully paid for the game. Have fun!
Dom Seeker, you have successfully paid for the game. Have fun!
Customer has successfully been added to the system
Paige Barclay, you have successfully paid for the game. Have fun!
Dom Seeker, you don't have enough credits
Paige Barclay, you don't have enough credits
Charles James-Warne, you don't have enough credits
Lucille Swan, you have successfully paid for the game. Have fun!
You are too young
Charles James-Warne, you don't have enough credits
Amber Y Basse, you have successfully paid for the game. Have fun!
Lucille Swan, you have successfully paid for the game. Have fun!
Lucille Swan, you have successfully paid for the game. Have fun!
Customer has successfully been added to the system
Amber Y Basse, you have successfully paid for the game. Have fun!
Dom Seeker, you don't have enough credits
Aoife Crouch, you don't have enough credits
Cat Chives-Keller, you have successfully paid for the game. Have fun!
Dom Seeker, you have successfully paid for the game. Have fun!
Charis Jaramillo, you have successfully paid for the game. Have fun!
Charis Jaramillo, you have successfully paid for the game. Have fun!
Cat Chives-Keller, you don't have enough credits
Customer has successfully been added to the system
Wrong format of id: id should contain 10 characters and start from letter A or C
 depends on the game!
Aoife Crouch, you have successfully paid for the game. Have fun!
You have successfully added money to your balance
Dom Seeker, you have successfully paid for the game. Have fun!
Paige Barclay, you have successfully paid for the game. Have fun!
Amanda Pham, you don't have enough credits
You have successfully added money to your balance
Amanda Pham, you have successfully paid for the game. Have fun!
Paige Barclay, you don't have enough credits
There is no customer with provided id
Failed to add funds
You have successfully added money to your balance
```

```
Paige Barclay, you have successfully paid for the game. Have fun!
Dom Seeker, you don't have enough credits
Dom Seeker, you don't have enough credits
Aman Crosby, you have successfully paid for the game. Have fun!
You have successfully added money to your balance
Amanda Pham, you have successfully paid for the game. Have fun!
Aman Crosby, you don't have enough credits
Amanda Pham, you have successfully paid for the game. Have fun!
Amanda Pham, you have successfully paid for the game. Have fun!
You have successfully added money to your balance
Dom Seeker, you have successfully paid for the game. Have fun!
Dom Seeker, you have successfully paid for the game. Have fun!
Charis Jaramillo, you have successfully paid for the game. Have fun!
Katriona Blownes, you have successfully paid for the game. Have fun!
Charis Jaramillo, you have successfully paid for the game. Have fun!
Customer has successfully been added to the system
Paige Barclay, you don't have enough credits
Lucille Swan, you have successfully paid for the game. Have fun!
Kiri Kramer, you have successfully paid for the game. Have fun!
Kiri Kramer, you have successfully paid for the game. Have fun!
Aoife Crouch, you don't have enough credits
Paige Barclay, you have successfully paid for the game. Have fun!
Kiri Kramer, you have successfully paid for the game. Have fun!
Charis Jaramillo, you have successfully paid for the game. Have fun!
Kiri Kramer, you have successfully paid for the game. Have fun!

The richest customer is

Name: Mantis Toboggan
Id: B473Z4
Age: 72
Level of discount: STAFF
Current balance: 20100

The median game price is: 200

Cabinet/Active/Virtual games: [7, 8, 7]
GamesCo does not take responsibility for any accidents or fits of rage that occu
r on the premises

Arcade name: Arcade of fun
Revenue:  8775

Games list: [

Name: "Virtual UEA Tour"
Id: AVI1USPBNG
Price: 0p
Equipment type: HEADSET_ONLY,

Name: "Reaction Test 2000"
Id: CBGCR27FQM
Price: 40p
Status of the reward: : true,

Name: "D'Arcy Thompson Pinball"
Id: CPVOHUHOZH
Price: 50p
Status of the reward: : true,

Name: "Foosball"
Id: AHWOHK1FO3
Price: 80p
```

```
Minimum age:3,

Name: "American Pool"
Id: A4FJTZLIVA
Price: 100p
Minimum age:12,

Name: "CMP Chomp Man"
Id: C7S6SYBL8R
Price: 120p
Status of the reward: : false,

Name: "Snooker"
Id: AJFS153KZV
Price: 120p
Minimum age:12,

Name: "Street Wrestler V"
Id: CQLS3YESOM
Price: 140p
Status of the reward: : false,

Name: "Air Hockey"
Id: AD65E3UJQJ
Price: 180p
Minimum age:12,

Name: "Darts"
Id: AX5YNVUJA9
Price: 200p
Minimum age:16,

Name: "Plumber Kart 8"
Id: CNQZPI7G5E
Price: 200p
Status of the reward: : false,

Name: "Plonky Kong"
Id: CAPSD7TLC6
Price: 200p
Status of the reward: : false,

Name: "Clock Crisis"
Id: CXPVCODBXU
Price: 220p
Status of the reward: : true,

Name: "Fly Like a Seagull!"
Id: AV55GWU6PS
Price: 350p
Equipment type: HEADSET_ONLY,

Name: "Virtual Petting Zoo"
Id: AVLD1ZDNXE
Price: 400p
Equipment type: HEADSET_AND_CONTROLLER,

Name: "Snowball Fight"
Id: AVX5KN5T3O
Price: 400p
Equipment type: BODY_TRACKING_SUIT,

Name: "Football Pool"
```

```
Id: AMURG8FXMK
Price: 500p
Minimum age:3,

Name: "Pickaxe Crafting Simulator"
Id: AVPO9GJA1Z
Price: 500p
Equipment type: HEADSET_AND_CONTROLLER,

Name: "Dance like a Professor"
Id: AVSKVMRB9U
Price: 800p
Equipment type: BODY_TRACKING_SUIT,

Name: "Beer Pong"
Id: AL2ETWHG0Q
Price: 1000p
Minimum age:18,

Name: "VR Paintball"
Id: AV9OPS1LRT
Price: 1000p
Equipment type: BODY_TRACKING_SUIT,

Name: "Axe Throwing"
Id: AN234FQD9D
Price: 2000p
Minimum age:18]

List of customers: [

Name: Barry Guy
Id: 1C6498
Age: 17
Level of discount: NONE
Current balance: 0,

Name: Charles James-Warne
Id: 51CX13
Age: 12
Level of discount: NONE
Current balance: 10,

Name: Jacob Peraltera
Id: P54578
Age: 32
Level of discount: STUDENT
Current balance: 10,

Name: Amanda Pham
Id: A64248
Age: 45
Level of discount: NONE
Current balance: 40,

Name: Edwin Shea
Id: 144374
Age: 24
Level of discount: NONE
Current balance: 100,

Name: Penelope Nootly
Id: 80O813
```

```
Age: 5
Level of discount: NONE
Current balance: 100,

Name: Dom Seeker
Id: 174450
Age: 19
Level of discount: STUDENT
Current balance: 108,

Name: Aman Crosby
Id: 546R82
Age: 62
Level of discount: STAFF
Current balance: 161,

Name: Lemmington Steele
Id: 889027
Age: 18
Level of discount: STUDENT
Current balance: 200,

Name: Jorge Pimento
Id: 985F83
Age: 23
Level of discount: STUDENT
Current balance: 200,

Name: Katriona Blownes
Id: 681212
Age: 21
Level of discount: NONE
Current balance: 200,

Name: Irene Freeman
Id: 387036
Age: 10
Level of discount: NONE
Current balance: 240,

Name: Aurora Mcleod
Id: 9F0610
Age: 39
Level of discount: STAFF
Current balance: 260,

Name: Paige Barclay
Id: 901420
Age: 29
Level of discount: NONE
Current balance: 276,

Name: Cat Chives-Keller
Id: 506VX5
Age: 15
Level of discount: NONE
Current balance: 280,

Name: Lois Romero
Id: 203685
Age: 41
Level of discount: STAFF
Current balance: 360,
```

```
Name: John-Paul Clay
Id: 58R526
Age: 47
Level of discount: STAFF
Current balance: 400,

Name: Maaria Crossley
Id: 555765
Age: 39
Level of discount: STAFF
Current balance: 400,

Name: Charis Jaramillo
Id: 518370
Age: 19
Level of discount: STUDENT
Current balance: 558,

Name: Lucille Swan
Id: 748A66
Age: 31
Level of discount: STAFF
Current balance: 602,

Name: Thea Worthington
Id: SGRQ01
Age: 17
Level of discount: NONE
Current balance: 800,

Name: Roshni Stephens
Id: 117DA4
Age: 15
Level of discount: NONE
Current balance: 1000,

Name: Moses Rivers
Id: 54A0N3
Age: 20
Level of discount: STUDENT
Current balance: 1000,

Name: Khloe Stokes
Id: 522136
Age: 6
Level of discount: NONE
Current balance: 1000,

Name: Aoife Crouch
Id: 738164
Age: 29
Level of discount: NONE
Current balance: 1000,

Name: Jimmy Baker
Id: 973SA2
Age: 4
Level of discount: NONE
Current balance: 1000,

Name: Amber Y Basse
Id: 6048D1
```

```
Age: 37
Level of discount: STUDENT
Current balance: 1020,

Name: Nour Buxton
Id: 343172
Age: 5
Level of discount: NONE
Current balance: 1200,

Name: Rowena Kaiser
Id: 264474
Age: 19
Level of discount: STUDENT
Current balance: 1600,

Name: Kiri Kramer
Id: 2612CX
Age: 38
Level of discount: NONE
Current balance: 2400,

Name: Lewie King
Id: 305459
Age: 47
Level of discount: NONE
Current balance: 3200,

Name: Amy Mckinney
Id: 3S6965
Age: 17
Level of discount: NONE
Current balance: 4000,

Name: Gravin Clawley
Id: 18761S
Age: 7
Level of discount: NONE
Current balance: 5000,

Name: Mantis Toboggan
Id: B473Z4
Age: 72
Level of discount: STAFF
Current balance: 20100]
```

Messages to STDERR

None.